How Much of DeFi Liquidity Is Higher-Order? Survivorship, LP Representation, and a Null That Holds

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Abstract

How much of the loop structure of a DeFi liquidity network is genuinely higher-order, and is that quantity stable? Curve's 3pool binds {USDC, USDT, DAI} in a single contract, a three-way relationship a pairwise graph discards and the *nerve complex* of pool co-membership keeps. Measured over Ethereum stablecoin pools by persistent homology, the quantity is **not stable at all**. The higher-order fraction swings from 0.31 to 0.93 across two defensible modeling choices that are rarely reported as sensitivities. The larger driver is **LP-token representation**. Resolving an LP token such as 3CRV into its base assets collapses metapools toward a cone and roughly doubles the fraction; keeping 3CRV as its own vertex does not. The second driver is **survivorship**, and it is fixable. Standard registry-based reconstruction filters today's pool list backwards, silently conditioning on survival, and recovers only ∼10 % of the loop structure. Internet Archive snapshots of the registry endpoint recover the rest: we rebuild the full historical universe over 36 snapshots (Oct 2022 to May 2025). De-censoring also revives the LP-resolution comparison, untestable on survivors alone since those metapools have all delisted. Against that fragility, one result survives every choice we vary. Through both the March 2023 USDC depeg and the July 2023 Curve/Vyper exploit the reconstructed structure is **inert**. An event window is indistinguishable from an arbitrary window of equal length, under either representation and on either sample. We bound that null rather than assert it. Synthetic damage shows the measure tracks how many pools die, not how much value: a shock had to eliminate about a fifth of the universe to register, and the depeg delisted 15 pools holding 0.8 % of TVL. An apparent signal at the exploit traced to corrupted TVL in the one archive crawl captured mid-event, and repairing it removes the signal. That is a second data-quality trap beside survivorship, and registry-based work has to guard against it. We read the null economically: stablecoins are common-numéraire substitutes, so their liquidity scaffold is shallow and does not re-wire under price stress. The methodological lesson is to report the higher-order content of a DeFi network as a sensitivity range over survivorship and representation, never as a point estimate. Code and data are fully reproducible.

Keywords Decentralised finance · Higher-order networks · Persistent homology · Survivorship bias · Automated market makers

JEL classification C81 · D85 · C65 · E42 · G14 · G23

1 Introduction

Classical financial-contagion models are pairwise: institutions are nodes, bilateral exposures are edges (Eisenberg & Noe, 2001). A multi-asset AMM pool is not. Curve's 3pool binds {USDC, USDT, DAI} into one contract, and a graph records only the three pairwise projections, discarding the joint membership. Higher-order network methods (hypergraphs, simplicial complexes; Bick et al., 2023) retain exactly that information, and a nascent literature argues they matter for systemic risk (Forte, 2026). The decisive reason to test this in decentralised finance rather than traditional finance is **observability**: pool compositions and reserves are public and machine-readable on-chain, removing the confidentiality barrier that has always constrained structural contagion research.

We make "how much is higher-order?" a measurement. For each snapshot we build the nerve complex of pool co-membership and compute the *skeleton-vs-nerve decomposition*: the cycle rank of the 1-skeleton ($E - V + B_0$, the loops a graph sees) vs. the Betti-1 number B_1 of the nerve (loops surviving once multi-asset pools fill their faces). Their ratio is the fraction of loop structure that is genuinely higher-order.

The central finding is that this fraction is **not a property of the network but of the analyst's choices**. Two knobs that any such study must set, and that are rarely reported as sensitivities, move it from a clear minority to an overwhelming majority:

1. **LP-token representation.** A Curve metapool reads nominally as $\{X, 3\text{CRV}\}$. Resolving 3CRV into its underlying {USDC, USDT, DAI} makes every such pool a simplex containing the 3pool triangle as a face, pushing the complex toward a cone (contractible, higher-order by construction); keeping 3CRV as its own vertex does not. This choice is the *larger* driver.
2. **Survivorship.** The standard historical reconstruction filters *today's* registry back in time, so it silently conditions on pools that survived, disproportionately the large multi-asset pools. This inflates the higher-order fraction and, we show, is fixable.

Our contributions:

- **A de-censoring method for DeFi registry survivorship** (§3): Internet-Archive snapshots of the registry recover the full historical universe, delisted pools included. Survivor-only reconstruction captures ∼1/10 of the loops (§5.1); the method applies to any registry-based DeFi network study.
- **A sensitivity decomposition of the higher-order fraction** (§5.2): it spans 0.31–0.93 across representation × survivorship; we separate the two contributions and show even the *sign of the time-trend* is convention-dependent. De-censoring revives the otherwise-untestable LP-resolution fork.
- **A convention-independent null on two events** (§5.3): the reconstructed structure is inert through *both* the USDC depeg and the Curve/Vyper exploit under every

choice we vary. It is the one representation- and sample-independent structural statement the data supports. Establishing it required a second data-quality guard: an integrity scan plus transient-dip repair that removes a spurious signal from the corrupted archive crawl captured mid-exploit.

A rigorous negative on the dynamics, and a demonstration that the descriptive metric is an artifact, together map where higher-order topology does and does not add value over pairwise methods, which we did not know in advance.

2 Related work

Higher-order finance is early-stage. Forte (2026) give the first hypergraph analysis of bank–firm credit networks (Central Bank of Argentina registry, adjusted H-eigenvector centrality): TradFi credit and centrality, not persistent homology and not DeFi. **Return-space TDA** is by contrast saturated: persistent homology of return point clouds covers equities (Gidea & Katz, 2018), FX, and crypto (de Favereau de Jeneret & Diamantis, 2025); we keep this lineage strictly separate, as our complex is built from *structure* (co-membership), not return correlation. **Graph analysis of DeFi** exists: Yan & Tessone (2025) model Uniswap as a token-node/pool-edge network, exactly the pairwise baseline our decomposition sits atop; Aufiero et al. (2025) survey the surrounding systemic-risk literature. To our knowledge the persistent-homology / nerve treatment of DeFi liquidity structure is previously unoccupied, and, more to the point, the survivorship and representation sensitivities we quantify are not addressed in registry-based DeFi network work, which typically reconstructs history from the current registry without either control.

3 Data: the registry, and de-censoring its survivorship

We use two free, unauthenticated DeFiLlama endpoints: the pool registry (`yields.llama.fi/pools`: project, chain, symbol, `underlyingTokens`, current TVL, stablecoin flag) and per-pool daily TVL history (`yields.llama.fi/chart/{pool}`). The universe is Ethereum, stablecoin-flagged pools with 2–8 underlying tokens.

The standard reconstruction, and its hidden conditioning. Since a pool's token set is fixed at deployment, the usual historical complex at date t is today's registry filtered to pools with TVL above threshold at t. This silently conditions on survival: pools delisted before today are invisible. It has been treated (including in two independent reviews of an earlier draft of this work) as an *unrecoverable* ceiling.

De-censoring. It is not unrecoverable. The Wayback Machine archived the registry endpoint roughly weekly from October 2022; each snapshot is the *full* universe at that date, delisted pools included, with the identical schema. We fetch 42 archived crawls (Oct 2022–May 2025, monthly, with tight clusters around the USDC depeg and the July 2023 Curve/Vyper exploit); 36 of these form the primary monthly series and the remaining six are the event clusters. We rebuild the complex on the full historical universe, and, per snapshot, also restrict to pools surviving to today, isolating the survivorship effect at fixed representation. True survivorship rises from 24 % to 41 % over the period (mechanically: recent pools are likelier to persist); at the depeg only 66 of 274 universe pools survive, and the 208 invisible pools were 35 % of universe TVL (FRAX-3CRV \$474M, MIM/OUSD/TUSD/LUSD-3CRV, ...). Terra (May 2022) predates the archive and is not de-censorable; snapshots are weekly crawl dates, not daily. One caveat we make operational below: a crawl captured during an acute event can carry corrupted TVL, so snapshots are run through an integrity scan (§5.3).

4 Method

Complex and filtration. Vertices are tokens (by contract address); a pool with token set σ contributes the filled simplex on σ (downward closure). We filter by each pool's *share* of universe TVL at date t, entering strong pools first at $f = -\log_{10}(\text{share})$, a level-invariant weighting, so a result cannot be the mechanical TVL drawdown of a depeg in disguise. We track B_0, essential B_1, B_2 ($\equiv 0$ throughout), the 1-skeleton cycle rank $E - V + B_0$, and the higher-order fraction $\text{HO} = (E - V + B_0 - B_1)/(E - V + B_0)$. Topology code is validated on hand-checkable toy complexes (a hollow triangle has one loop, a filled one none, a tetrahedron boundary has $B_2 = 1$).

Why essential B_1 and not persistence summaries. The filtration above supports the standard return-space TDA summaries (persistence landscapes, L^p norms, total persistence) and we do not use them, for a reason worth stating. On this object almost every H_1 class is essential: across all 42 crawls under both representations, 35 of 3430 H_1 classes have finite persistence (1.0%), never more than two on a single crawl against 40–58 essential ones; on the daily survivor-only series the rate is exactly zero across 579 consecutive days. Under the discard-infinite convention that `gudhi` and the return-space pipeline use by default, the H_1 landscape is therefore identically zero or near enough. Capping infinite bars instead makes the norms non-zero, but they then track the arbitrary cutoff rather than the data, and event and placebo windows stay indistinguishable (L^2 1.22 vs. 1.25). H_0 is the control: it has genuine finite bars and the machinery works there. The reason is structural, not numerical — a loop here closes only if some pool contains all of its tokens, which weaker pools entering later do not do — so the failure is specific to H_1 on a co-membership complex. We read this as a transferable caution: the return-space TDA toolkit does not carry to structural liquidity complexes, and Betti numbers are the appropriate summary.

The two modeling axes. We cross **sample** (survivor-only vs. de-censored/full) with **LP representation**: *lp_vertex* keeps LP tokens as their own vertices (the archive's native form), *resolved* replaces the 3CRV LP token (identified as `0x6c3f...e490`, present in ~68 metapools per snapshot in the archive) with $\{\text{DAI}, \text{USDC}, \text{USDT}\}$. Today's API base-resolves, so the survivor reconstruction is implicitly the *resolved* fork; the archive's native form makes *lp_vertex*, and with it the comparison, available for the first time.

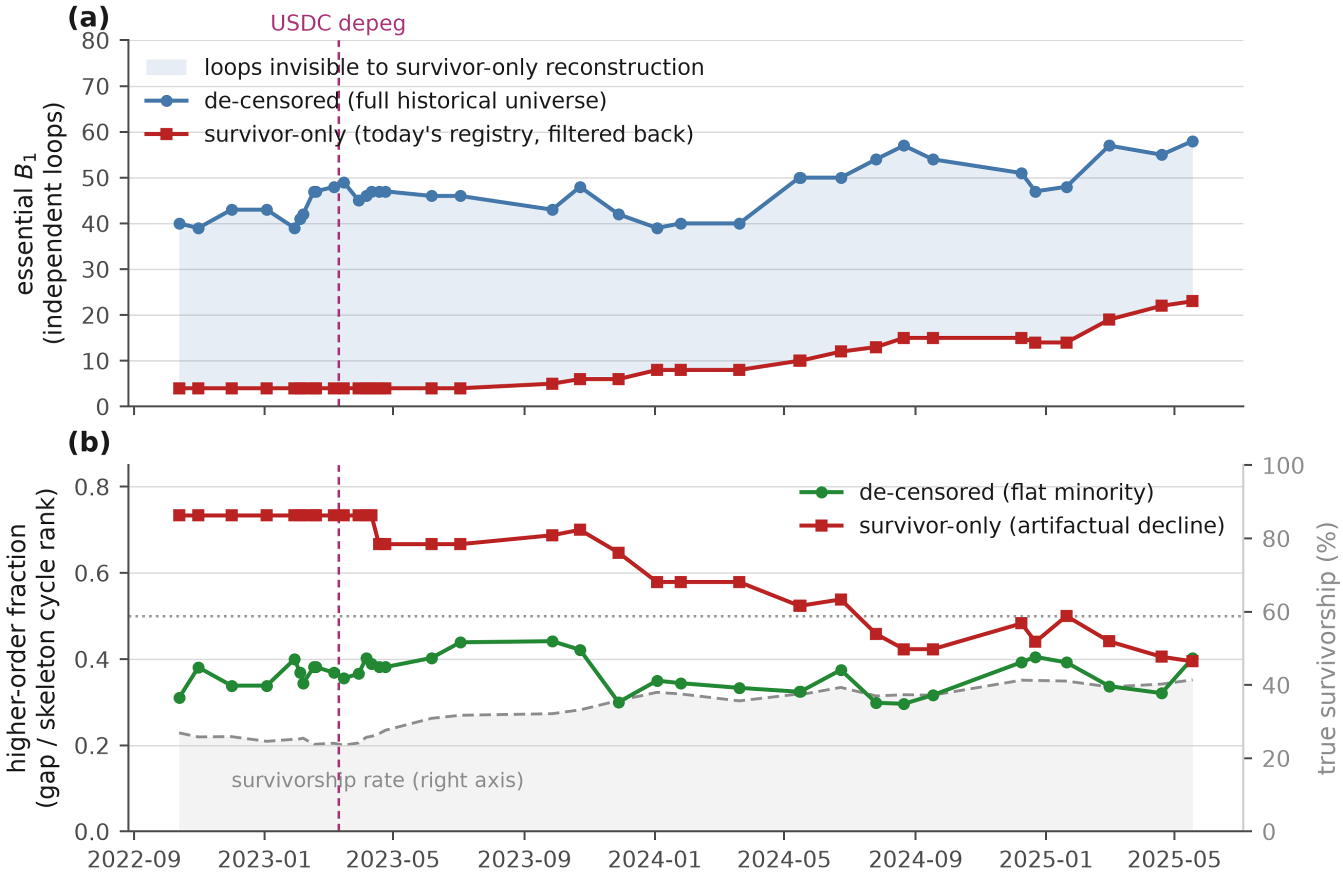

Fig. 1 Survivor-only reconstruction sees roughly one loop in ten, and manufactures a trend that is not there. **a** Essential B_1 on the de-censored full historical universe (blue) vs. on pools that survive to today (red); the shaded wedge is the loop structure a survivor-only study cannot see. **b** The higher-order fraction is flat on the full universe (green) but declines on survivors (red), and that decline tracks the rising survivorship rate (grey, right axis): it is the survivor sample converging to the truth, not the network changing. Vertical line: the USDC depeg (2023-03-11).

Event test and data-quality guard. With snapshots at weekly cadence we test each event non-parametrically: is the pre→post change large relative to the distribution of all consecutive-snapshot changes across the series? We run two events (the USDC depeg, 2023-03-07 → 03-16; the Curve/Vyper exploit, 2023-07-30 → 08-02) under both representations. Since a crawl captured mid-event can be corrupted, every snapshot first passes an *integrity scan* (flag if universe TVL is $< 0.6\times$ the median of its temporal neighbours); a flagged snapshot's *transient dips* (pools materially live on both neighbours but $< 0.5\times$ the smaller on the flagged crawl) are repaired by neighbour interpolation, the snapshot-level analogue of the daily gap-fill. (An earlier daily survivor-only analysis using a placebo-window permutation test over 632 daily complexes reaches the same null; we report the de-censored version as primary evidence.)

5 Results

5.1 Survivor reconstruction sees ~1/10 of the loop structure

At fixed representation (lp_vertex), essential B_1 is **40–58 on the full universe vs. 4–23 on survivors**, at every snapshot; the true universe is 3–4× the survivor universe (234–430 vs. 63–178 pools). The object a survivor-only study analyses is a small, biased slice, and the bias is toward exactly the large multi-asset pools that dominate the higher-order fraction.

5.2 The higher-order fraction is a modeling artifact

Table 1 crosses the two axes. The higher-order fraction spans **0.31–0.93**. Reading it two ways:

- **Representation is the larger driver.** Resolving 3CRV collapses ~68 metapools onto a shared base triangle (a partial cone): across 36 snapshots it removes ~28 % of loops (essential B_1 46.8 → 33.8) yet nearly *doubles* the higher-order fraction (0.36 → 0.67). At the depeg, full-universe HO is 0.37 (lp_vertex) vs. 0.75 (resolved).
- **Survivorship amplifies in the same direction.** Holding representation = resolved (the paper-standard convention), full→survivor pushes HO 0.75 → 0.93 at the depeg, since survivors are the big multi-asset pools.
- **Even the trend is convention-dependent** (Fig. 1). Under resolved, the de-censored fraction genuinely *declines* 0.80 → 0.50 as metapools delist; under lp_vertex it is *flat* at 0.36 ± 0.04. The survivor fraction declines under both, but that is the survivor sample converging to the full-universe value as survivorship rises (corr(survivor HO, survivorship) $= -0.92$ for lp_vertex), not a change in the network.

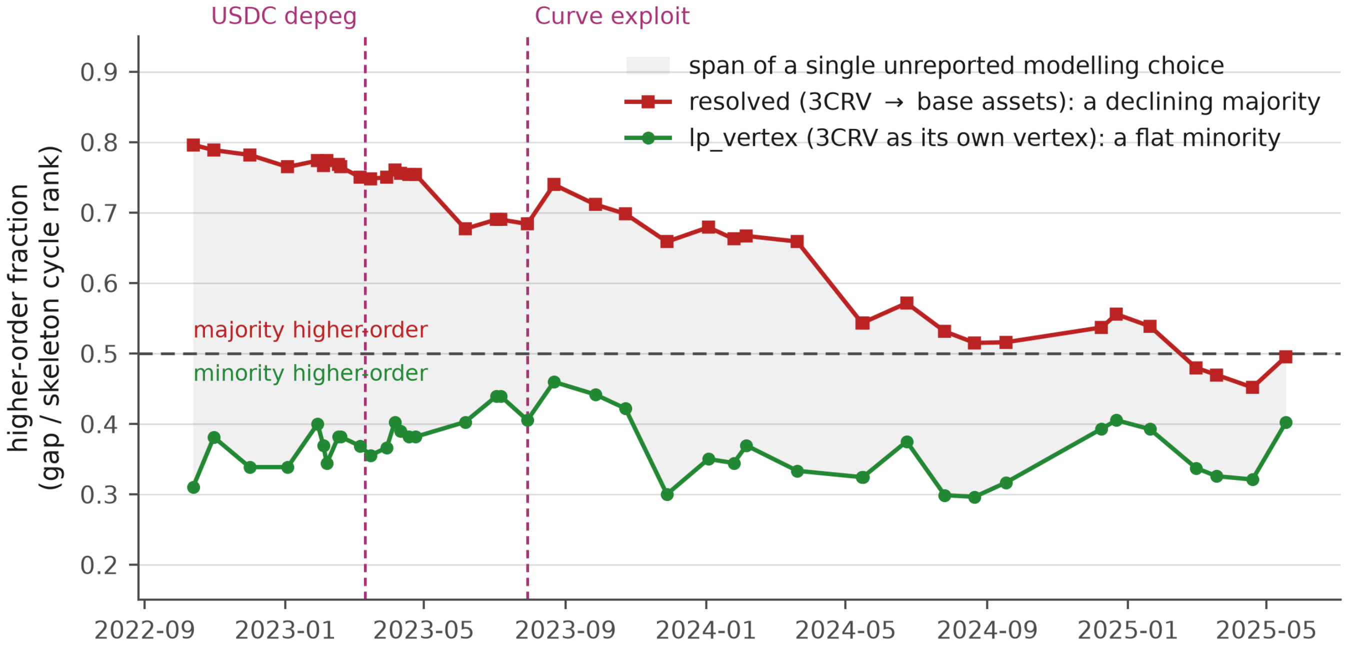

Fig. 2 The *same* de-censored universe reads as majority- or minority-higher-order purely by LP-token representation. Resolving 3CRV to base assets (red squares) keeps the fraction a declining majority (> 0.5); keeping 3CRV as its own vertex (green circles) keeps it a flat minority (< 0.5). The shaded band between them is the span of one modelling choice that is rarely reported; the two conventions sit on opposite sides of the majority line for the entire period, converging only as the metapool structure dissolves. The corrupted 2023-08-02 crawl (Sec. 5.3) is excluded.

Table 1 Higher-order fraction by sample $\times$ LP representation. The paper-standard corner (resolved, survivor) reads a declining majority; the de-censored, composability-preserving corner (lp_vertex, full) reads a flat minority. Same data.

snapshot	lp_vertex, full	lp_vertex, survivor	resolved, full	resolved, survivor
2022-10	0.31	0.73	0.80	0.93
2023-03	0.37	0.73	0.75	0.93
2023-06	0.40	0.67	0.68	0.92
2023-11	0.30	0.65	0.66	0.87
2024-05	0.32	0.52	0.54	0.77
2024-12	0.39	0.48	0.54	0.68
2025-05	0.40	0.39	0.50	0.60

So the paper-standard "a majority of loops are higher-order, declining over 2022–23" sits at the high-inflation corner; the de-censored, composability-preserving reading is a flat minority (Fig. 2). Both conventions are economically defensible (resolving reflects exposure to base assets; lp_vertex reflects composability structure), but they are not symmetric in kind. Resolving carries a mechanical inflation that lp_vertex has no counterpart for: each metapool becomes a filled simplex sharing the 3pool triangle, adding one vertex and three edges to a common base, so the skeleton cycle rank rises by two while B_1 does not move. The gap widens by construction, $\sim$68 times over. Under lp_vertex the same pool is a single edge on a star centred at 3CRV and adds nothing to either quantity. What lp_vertex costs is interpretive rather than mechanical: a holder of FRAX-3CRV is exposed to four assets and lp_vertex records a two-asset relationship. Only one of the two conventions manufactures the quantity being measured, which is worth saying plainly even though it does not pick a winner. **The honest answer to "how much is higher-order?" is the range in Table 1, not a number.**

5.3 The null is the part that holds

Against all of that fragility, the dynamics are stable, and stay stable on a second event and after data-quality control. Table 2 runs the de-censored event test on both events. Two details of the test matter enough to state. First, crawl spacing is irregular (1–83 days, median 26) while the event windows are 9 and 3 days, and drift grows with the gap — mean $|\Delta|$ in essential B_1 is 1.40 on intervals of $\leq$14 days against 2.44 across all intervals — so judging an event against the full pair list would bias the comparison toward the null it reports. We therefore match on spacing. Second, the observables are small integers, so $|\Delta|$ ties heavily and the percentile depends on whether ties are counted below or at the event value; on the depeg that choice alone moves essential B_1 from the 23rd to the 49th percentile. We use the mid-rank convention throughout. Under both corrections the events land near the *median* of comparable routine intervals rather than in the lower tail: the

Table 2 De-censored event test, full universe, both events. $|\Delta|$ is the change across the event window (depeg 2023-03-07 $\to$ 03-16, 9 days; exploit 2023-07-30 $\to$ 08-02, 3 days, after transient-dip repair). Percentiles are mid-rank against *gap-matched* reference intervals ($\leq$14 days, $n = 15$), not all 41 consecutive pairs, whose median spacing is 26 days and whose drift is correspondingly larger. "Detect" is the smallest $|\Delta|$ clearing the 95th percentile of that reference set: what the test could have seen, as against what it did. Both events sit near the middle of routine variation for comparable windows, and well under the threshold, under both representations.

observable	repr.	$\|\Delta\|$ depeg / expl.	pct.	detect
ess. B_1	lp_vertex	1 / 1	53rd	5
gap	lp_vertex	1 / 1	47th	7
ess. B_1	resolved	0 / 0	27th	4
gap	resolved	1 / 1	50th	11

sharper statement is that an event window is indistinguishable from an arbitrary window of the same length, not that it is unusually quiet.

A second event, and a data-quality trap. The Curve/Vyper exploit (2023-07-30) at first appears to break the null: on the de-censored universe essential B_1 jumps 47 $\to$ 55 (90th percentile) under lp_vertex and 31 $\to$ 37 under resolved. But the integrity scan flags the single 08-02 crawl as the *only* anomaly in the series (universe TVL halves, \$3.04B$\to$ \$1.73B, then recovers), and 84% of that drop is 10 transient-dipped registry rows (6 distinct pools; DeFiLlama re-lists the same Curve pool under Convex), 8 of them FRAX (FRAX-USDC \$435M$\to$ \$73M$\to$ \$451M; FRAX-USDP \$112M$\to$ \$3M$\to$ \$129M): corrupted readings during the exploit, not a real drawdown (the exploit drained \$70M of *ETH* pools, not stablecoins). Repairing the transient dips removes the signal entirely: essential B_1 returns to 48, its pre-event value (Fig. 3), and the Curve window is inert. The null therefore holds on $N = 2$ events, each confirmed only after snapshot data QC. The episode is itself a result: *archived registry snapshots crawled during acute events can manufacture spurious topological signals. It is a second data-quality trap beside survivorship, and one the naive analysis would have reported as the sought-after "topology detects the exploit" finding.*

Unlike the descriptive fraction, the null holds under every choice we can vary: survivorship, LP representation, sampling cadence, and now two distinct stress events. It directly refutes the natural objection that the survivor view looks rigid only through omitting the pools that broke: restoring all 208 dead pools, the structure is still inert.

5.4 How large a shock would have registered?

A null on an integer-valued observable at weekly-to-monthly cadence invites an obvious objection: "the structure did not move" and "this test cannot resolve a move this small" produce the same output. We separate them by damaging a real snapshot synthetically and locating where essential B_1 leaves the band of routine drift.

From the 2023-03-07 crawl we delete pools three ways: the N largest by TVL; largest-first until X% of universe TVL is gone; and a random pool set totalling the same X%, averaged over 12 seeds. Fig. 4 gives the dose-response against the threshold of 5 loops.

The first result is that **the observable responds to pool count, not to value**. Deleting the eight largest pools removes 54 % of universe TVL and moves essential B_1 by 3, under threshold; deleting a random fifth of TVL removes about 60 pools and moves it by 7. Targeted destruction stays under threshold to 70 % of TVL, while the diffuse threshold sits at 20 % on all four snapshots we repeated this on (2022-10, 2023-03, 2024-05, 2025-05). That is a property of the nerve: a loop needs distinct pools weaving tokens together, and a large pool contributes its own filled simplex as readily as its edges.

The second result **bounds the null**. Across the depeg window 15 pools delisted, holding 0.8 % of universe TVL — roughly an order of magnitude below detectability in the dimension the observable actually responds to. Universe TVL did fall 24 % over the window, but almost all of that was repricing and outflow inside pools that survived. So the reading in §6, that a depeg moves prices and flows within a fixed scaffold rather than re-wiring it, is not only an interpretation placed on a null: the injection test measures that mechanism directly, and the injection also supplies the positive control the two observed events cannot provide. The bound is specific to pool disappearance, which is the intervention a co-membership nerve is built to see; a shock that re-wired membership without delisting anything would need a different injection and we do not claim to bound it.

6 Discussion

The fragility and the null resolve under one economic reading. Stablecoins are near-perfect substitutes pegged to a common numéraire, so the liquidity network is a hub-and-spoke around "dollar-ness" (mostly the 3pool and shallow wrappers) rather than a rich web of distinct multi-way dependencies. Two consequences follow: (i) *how much* of that shallow scaffold counts as "higher-order" is genuinely under-determined: whether a metapool's dependence on the 3pool is one relationship (an LP vertex) or three (resolved base assets) is a modeling stance, not a fact, hence the 0.31–0.93 range; and (ii) a depeg is prices and flows moving *within* a fixed scaffold, not the scaffold re-wiring, hence the null.

The transferable lesson is methodological, and it is three traps deep. Registry-based reconstruction is the default in empirical DeFi network research; it silently conditions on survival (discarding ~90% of the loop structure), it hides an unremarked representation choice (which, with survivorship, can flip a headline descriptive claim), and, as testing the Curve exploit revealed, a crawl captured during an acute event can carry corrupted TVL that *manufactures* a topological signal. The last is the sharpest cautionary note: the naive pipeline would have reported the FRAX-glitch spike as the sought-after "topology detects the exploit" result. All three are cheap to address (de-censor via the Internet Archive, report the representation sensitivity, and run an integrity scan with transient-dip repair), but none is addressed by simply collecting more data. Where structural higher-order TDA might still

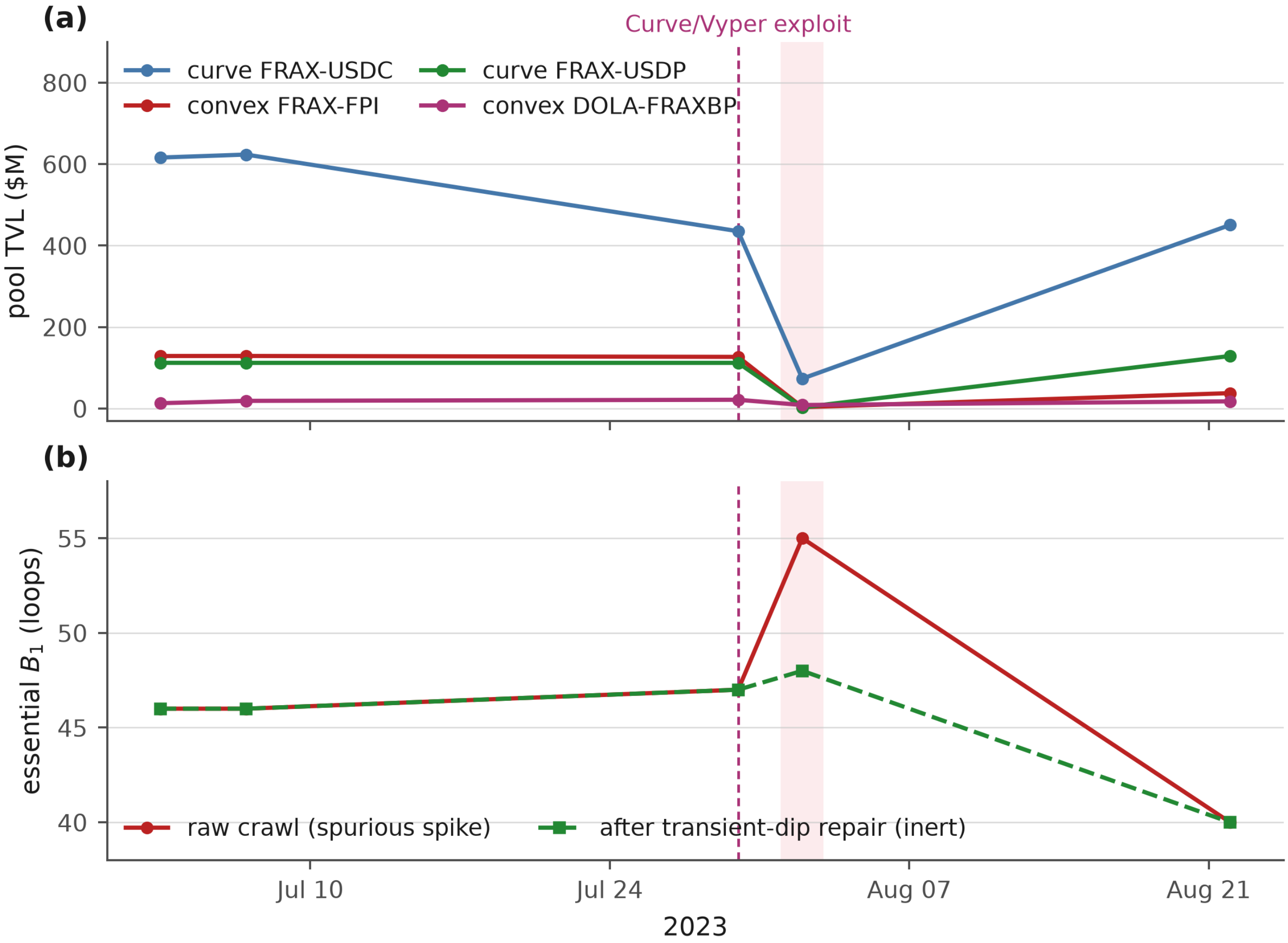

Fig. 3 The Curve-exploit "signal" is a data glitch. **a** The pools driving the apparent event crater to near-zero on the single 2023-08-02 crawl (shaded); three of the four rebound to at or above their pre-event level within three weeks (FRAX-USDC \$435M→\$73M→\$451M) and FRAX-FPI partially. A \$1.3B "drawdown" that reverses on the next crawl is a reading error, not a liquidity event. **b** That corruption injects a spurious essential-B_1 spike ($47 \to 55$, red) that transient-dip repair removes ($\to 48$, green), leaving the window inert. Dotted line: the exploit (2023-07-30); the 10 dipped registry rows are 6 distinct pools, 8 rows of them FRAX.

earn its keep is where these caveats bite less: non-substitute universes (mixed collateral, cross-asset lending) with heterogeneous multi-way dependence, and slow regime changes that actually re-wire membership.

7 Limitations

- **Two defensible conventions, no tie-breaker.** We report the range rather than adjudicate; a reader with a specific economic question (exposure vs. composability) should pick the matching convention and its number.

- **Two de-censorable events, and no observed positive control.** The USDC depeg and the Curve/Vyper exploit are both inside the archive window and both give the null, but both are episodes where §6 predicts membership should not re-wire, so on their own they cannot show that the measure would respond to anything. Terra (May 2022) predates the archive; the FTX collapse falls in a 31-day gap between crawls. The injection test (§5.4) supplies a synthetic positive control in place of an observed one, which is weaker: it establishes what *pool deletion* of a given size does, not what a real re-wiring event would do. Spacing limits the test directly, leaving 15 gap-matched reference intervals. (A daily survivor-only placebo permutation reaches the same null, but it is survivor-only, so §5.1 applies to it and we do not lean on it.)

- **One data source, and it demonstrably fails.** Every quantity here is DeFiLlama's, and §5.3 shows a single corrupted crawl nearly became the headline finding. The integrity scan and transient-dip repair catch the failure mode we found; they cannot certify the ones we did not. A registry error that is smooth across neighbouring crawls, or a systematic mis-listing rather than a transient dip, would pass both guards silently. Cross-checking TVL against an independent source (subgraph queries or direct chain reads) is the obvious next hardening, and until it is done, single-source risk is the largest uncosted exposure in the pipeline. A denser crawl cadence would also tighten the event tests, which are currently spacing-limited (Table 2).

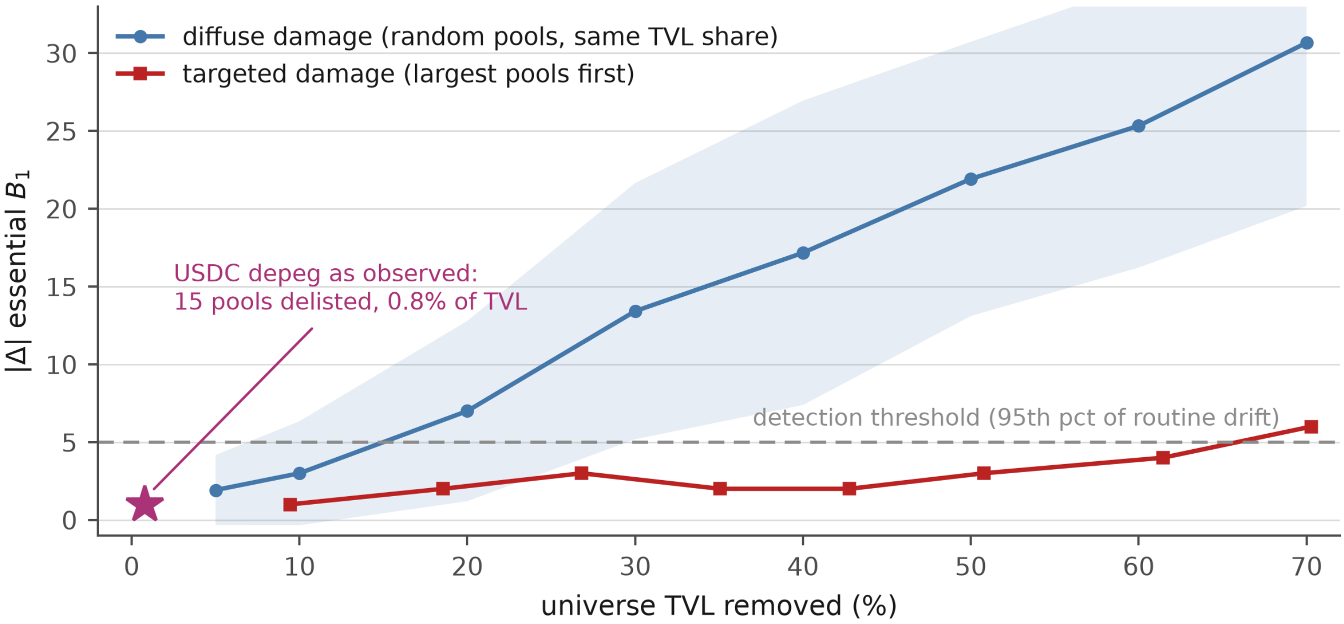

Fig. 4 What the event test could have seen. Starting from the depeg pre-event crawl (274 pools, \$5.73B), pools are deleted synthetically and the change in essential B_1 is recorded against the gap-matched detection threshold from Table 2 (dashed). Removing a random fifth of universe TVL, which costs about 60 pools, clears the threshold; removing the same value concentrated in the largest pools does not, even at 70%. The observable tracks how many pools are destroyed, not how much value. The star marks the depeg as it actually occurred: 15 pools delisted, holding 0.8% of universe TVL, roughly an order of magnitude below anything this test could resolve.

- **Ethereum stablecoins only.** The economic reading (common-numéraire substitutes) is exactly why the scaffold is shallow; other universes may differ, which is the point.
- **Cross-era representation.** The archive is natively lp_vertex and today's API resolved; we run both explicitly rather than mixing them.

8 Conclusion

Measured rather than asserted, the higher-order structure of Ethereum stablecoin liquidity has no convention-free size: the higher-order fraction ranges 0.31–0.93 across two modeling choices, LP-token representation (the larger) and survivorship, that registry-based DeFi network studies routinely leave unexamined, and even its time-trend flips sign between conventions. Survivor-only reconstruction, the field default, sees roughly a tenth of the loop structure; Internet-Archive de-censoring recovers the rest and revives an otherwise-untestable representation fork. The one convention-independent structural result is a null, and a bounded one: through both the largest stablecoin depeg on record and the Curve/Vyper exploit an event window looks like any other window of its length, and synthetic damage puts the detection floor near a fifth of the universe against a depeg that delisted 0.8 % of it. Reaching that conclusion required catching a third trap, a corrupted event-window crawl that would otherwise have read as a spurious detection. Report higher-order DeFi structure as a sensitivity range, de-censor the registry, scan for event-window data corruption, and treat the stable dynamics, not the fragile levels, as the finding.

Reproducibility

All code, the computed observable series, and figures are at `https://github.com/adamLEVi16/topology-engine` (`defi_topology/`). The pipeline runs from the free DeFiLlama API and the Internet Archive with no authentication; the raw registry snapshots (∼390 MB) are refetched rather than committed. Since the Wayback Machine is append-only, resolving a target date to its "closest crawl" can drift over time, so the exact snapshot timestamps used are recorded in the series JSON and the series is rebuilt byte-for-byte with `decensor.py --pinned` (each timestamp maps to a fixed, permanent crawl). Toy-complex tests pin the topology, and every number is regenerated by a single script per result (`decensor.py` for the de-censored series, the 2×2, the event tests, and the integrity scan / `repair_transient_dips` data-quality guard).

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